

Two Margins of International Track-and-Field Recruitment to NCAA Division I: Entry, Sublinear Scaling, and Event-Specific Heterogeneity

Alina Malkova
Florida Institute of Technology

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Abstract

We construct a country-level panel of international student-athletes in U.S. NCAA Division I rosters for the 2024–25 / 2025–26 season (track and field, $n = 18,657$; soccer, $n = 660$). The empirical setting is the international track-and-field market (96% of the international sample). We decompose athlete supply along two margins. **On the extensive margin** (whether a country sends any athletes), log GDP per capita and log population are strong independent predictors ($\hat{\beta} = 0.52$ and 0.18 respectively, both $p < 0.01$); political stability is not. **On the intensive margin** the headline fact is that athletes scale sublinearly with source-country population: estimated cohort elasticities are 0.33 in OLS on log-counts, 0.48 in a Negative Binomial, and 0.61 in a Heckman two-step that corrects for selection into the sender pool. Quantile regression sharpens the picture: the elasticity is 0.24 at the 25th percentile of supply and 0.51 at the 75th, indicating diminishing returns at low supply and roughly-proportional scaling at high supply. Political stability is statistically zero in OLS ($\hat{\beta} = 0.38$, $SE = 0.24$) but the data have only $\sim 80\%$ power to reject effects larger than 0.68 ; in count-data (NegBin) and selection-corrected (Heckman) specifications the same coefficient is $+0.79$ ($p < 0.01$) and $+0.60$ ($p < 0.10$). We therefore cannot make a strong statement on the intensive-margin role of governance; the data are consistent with a moderately positive effect. Within-event regressions reveal that the country-panel result averages over very different sport-specific structures: the Caribbean sprint pipeline and the East African distance-running pipeline respond to different economic primitives. An earlier draft of this work reported large institutional and disaster-event effects in a rate-model specification (athletes per cohort with log pop omitted); we show in Section 3.3 that those coefficients are artefacts of the rate model’s implicit unit-elasticity restriction, and we retract that framing. Macro covariates explain 21% of intensive-margin variation; eight regional fixed effects add another 11%; country-specific recruitment pipelines (Kenya–Iten, Jamaica–MVP, the Caribbean sprint network) plausibly account for the rest.

1 Data

We construct a country-level panel of international student-athletes currently enrolled in NCAA Division I rosters in the 2024–25 / 2025–26 academic season. Athlete-level records are scraped from official athletic-department websites for soccer (660 athletes from 27 schools) and track and field / cross country (18,657 athletes from 328 of 355 D-I programs; school coverage 92.4%). Only 68 of 1,830 international athletes compete in soccer, so we report the combined sample in headline tables

but characterise the empirical setting throughout as the *international track-and-field recruitment market*.

Hometown text is parsed from each roster page and mapped to ISO-3 country codes via a normalisation procedure that combines US state abbreviations (both postal and AP-style), Canadian provinces, World Bank country names, and a manual alias map for common roster forms. The mapping resolves 87.6% of track and 99.8% of soccer hometowns.

Country covariates come from World Bank WDI (GDP per capita PPP, total population, 15–24-year-old cohort 2018–2022 average, CPI inflation, merchandise terms of trade), Worldwide Governance Indicators (political stability, government effectiveness, rule of law, control of corruption), UCDP/PRIO conflict, EM-DAT natural disasters, the Oxford COVID-19 Government Response Tracker, and UNHCR Refugee Statistics. We use 2023 cross-sections and 2010–2023 cumulative aggregates. The intensive-margin analysis covers the **102 countries with at least one international athlete**; the extensive-margin probit uses a wider universe of 191 countries with non-missing covariates. We assign each country to one of eight regions for the regional fixed-effects analysis in Section ??: Sub-Saharan Africa, MENA, South Asia, East Asia & Pacific, Latin America & Caribbean, North America, Western Europe, and Eastern Europe & Central Asia.

Table 1. Descriptive statistics by income tier

Countries split into quartiles by 2023 GDP per capita PPP.

Tier (Q of GDP/cap PPP)	Countries	Athletes	Per country	GDP/cap PPP (med.)	Athletes/mil 15–24	Pol. stab
Q1 Low	26	533	20.5	\$7,670	0.84	
Q2 Lower-middle	25	109	4.4	\$21,917	2.38	
Q3 Upper-middle	25	352	14.1	\$41,288	12.15	
Q4 High	26	825	31.7	\$64,485	13.17	
All	102	1819	17.8	\$30,837	5.59	

2 Specification

We decompose international NCAA athlete supply into two margins. The extensive margin is whether country c sends any athletes,

$$\Pr(\text{any athletes}_c = 1) = \Phi(\alpha + \gamma_1 \log \text{GDPpc}_c + \gamma_2 \log \text{pop}_c + \gamma_3 \text{polstab}_c). \quad (1)$$

The intensive margin is the count of athletes among senders. We use $\log \text{pop}_{15-24,c}$ as a free regressor rather than a -1 offset:

$$\log \text{athletes}_c = \alpha + \beta_1 \log \text{pop}_{15-24,c} + \beta_2 \log \text{GDPpc}_c + \beta_3 \text{polstab}_c + \mathbf{x}'_c \boldsymbol{\delta} + \varepsilon_c. \quad (2)$$

Section 3.3 documents why a rate outcome (athletes/cohort) with log pop omitted gives misleading results. We complement OLS with (i) a Heckman two-step selection correction using the probit from Equation ??, (ii) a Negative Binomial specification on raw counts, which handles low-count truncation that OLS on log athletes does not, and (iii) quantile regression at $\tau \in \{0.25, 0.50, 0.75\}$. We report Huber–White (HC1) robust standard errors for OLS.

3 Results

3.1 Extensive margin

Probit: Pr(sends any athletes)	
log GDP per capita PPP	+ 0.52*** (0.11)
log population	+ 0.18*** (0.06)
Political stability	+0.19 (0.17)
n	191
Pseudo R^2	0.19

GDP per capita and population determine entry; political stability does not. Marginal effect of log GDPpc on Pr(sends) is approximately +0.20 per log-unit at the sample mean.

3.2 Intensive margin: three specifications

Different estimators give different answers on the intensive margin. We report all three side by side.

Coefficient	OLS log-count	Negative Binomial	Heckman 2-step (w/ IMR)
log pop _{15–24}	+ 0.33*** (0.11)	+ 0.48*** (0.15)	+ 0.61** (0.29)
log GDPpc	+ 0.37** (0.16)	−0.07 (0.18)	+1.44 (0.93)
Political stability	+0.37 (0.25)	+ 0.79*** (0.25)	+0.60* (0.34)
log(1 + disaster events)	+0.03 (0.19)	−0.01 (0.26)	+0.10 (0.19)
Inverse Mills ratio	—	—	+3.27 (2.64)
n	100	100	100
R^2 / pseudo- R^2	0.21	—	0.22

The sublinear-scaling fact is robust across estimators. The cohort population elasticity ranges from 0.33 (OLS) to 0.61 (Heckman). All three estimates are well below the unit elasticity that a constant-returns supply curve would predict. The Heckman estimate, which corrects for selection into the sender pool, sits in the middle of the reviewer-flagged range (0.5–0.7) and is our preferred point estimate. The substantive conclusion is the same across methods: doubling the source-country 15–24 cohort raises NCAA athlete supply by considerably less than a factor of two.

Political stability is not pinned down. The OLS coefficient (+0.37, SE 0.24) is statistically zero, but the data have only $\sim 80\%$ power to reject effects larger than $MDE = 1.96 \cdot 0.24 + 0.84 \cdot 0.24 \approx 0.68$ (?). The NegBin estimate is +0.79 ($p < 0.01$); the Heckman estimate is +0.60 ($p < 0.10$). Both are near the OLS MDE. The most defensible reading is that we cannot distinguish between zero and a moderately positive effect of about +0.7; the OLS-zero, NB-positive, and Heckman-marginal estimates are all within sampling variability of one another.

Log GDP per capita is fragile across estimators. OLS gives +0.37 ($p < 0.05$); NegBin gives −0.07 (n.s.); Heckman gives +1.44 with SE 0.93 (n.s., but the point estimate is large). Wealth is unambiguously important on the extensive margin (Section ??); on the intensive margin its role is estimation-method-dependent and we do not claim a precise effect.

Disaster events do not survive any specification. The coefficient is essentially zero in OLS, NegBin, and Heckman. The previous draft’s −0.74*** in the rate model was an artefact (Section ??).

3.3 Methodological note: the rate-model artefact

A previous version of this paper used $\log(\text{athletes}/\text{pop}_{15-24})$ as the intensive-margin outcome and *omitted* $\log \text{pop}$ from the right-hand side. That specification implicitly imposes a unit elasticity of athletes wrt population. The data sharply reject this restriction: the free-elasticity estimate is between 0.33 and 0.61 depending on the estimator, not 1.0. The mis-specification gets loaded onto the other coefficients.

Coefficient	Rate model (implicit $\beta_{\text{pop}} = 1$)	Count model (β_{pop} free, OLS)	Difference
$\log \text{GDPpc}$	+0.25 (0.22)	+0.37** (0.16)	—
Political stability	+1.12*** (0.27)	+0.37 (0.25)	collapses (OLS)
$\log(1 + \text{disaster events})$	−0.74*** (0.16)	+0.03 (0.19)	collapses
$\log \text{pop}_{15-24}$	(constrained = −1)	+0.33*** (0.11)	—
R^2	0.53	0.21	—

Small countries (Norway, Iceland, Switzerland, Bahamas, Jamaica) are systematically high on the WGI political-stability scale and have relatively few EM-DAT disaster events; large countries (India, China, Indonesia, Pakistan) have the opposite. When the rate model constrains the population elasticity to −1, this small-vs-large correlation loads onto the stability and disaster coefficients. The collapse from +1.12 to +0.37 in OLS—and to +0.79 in NegBin or +0.60 in Heckman—is what the artefact correction looks like.

We retract the previous draft’s headline that “political stability dominates the intensive margin” and replace it with the more cautious reading in Section ??: stability may matter, but our identification of its magnitude is loose.

3.4 Quantile regression: heterogeneity in the population elasticity

OLS estimates the conditional mean; if the country-pipeline residual story is right, the population-supply relationship may differ across the distribution of athlete supply.

Coefficient	$\tau = 0.25$	$\tau = 0.50$	$\tau = 0.75$
$\log \text{pop}_{15-24}$	+0.24**	+0.42***	+0.51***
$\log \text{GDPpc}$	+0.28	+0.49*	+0.39
Political stability	+0.21	+0.46	+0.78*

The population elasticity **more than doubles** from $\tau = 0.25$ to $\tau = 0.75$ ($0.24 \rightarrow 0.51$). Among countries supplying few international athletes per cohort (the lower quartile), the relationship is very weak; among high-supply countries (the upper quartile), the relationship is closer to proportional. The OLS average of 0.33 smooths these together and probably understates both the floor and the ceiling. Political stability also moves strongly across the distribution (+0.21 at $\tau = 0.25$ to +0.78 at $\tau = 0.75$), suggesting institutional quality matters mostly for countries already supplying many athletes, where the marginal unit of governance can extend the existing pipeline rather than build one from scratch.

3.5 Sport-event heterogeneity

The international concentration is in track. We exploit the position-field information in our scraped roster data to assign each athlete to a track event group (Sprints, Distance, Throws, Jumps) and re-estimate the count model within each group. This is the cleanest way to test whether the country-panel result is masking sport-specific structure.

Coefficient	Sprints	Distance	Throws	Jumps
$\log \text{pop}_{15-24}$	+0.13**	+0.40***	+0.18**	+0.10
$\log \text{GDPpc}$	+0.05	-0.46	-0.11	-0.05
Political stability	+0.13	+0.93**	+0.31	+0.06
n	43	40	45	42
R^2	0.07	0.23	0.10	0.06

The sport-event split reveals two qualitatively different country structures.

Sprints look Caribbean. Small population elasticity (+0.13, marginally significant), zero GDP coefficient, zero political-stability coefficient. The supply comes disproportionately from small high-quality pipeline countries (Jamaica, Bahamas, Trinidad, Barbados) that move little with the macro covariates because they are clustered tightly together on those covariates and differentiated by network structures the macro model does not see.

Distance running looks East African. Strong population elasticity (+0.40), *negative* (insignificant) GDP coefficient, and a large positive political-stability coefficient (+0.93, $p < 0.05$). The negative GDP sign reflects the East African concentration: distance running is dominated by Kenya, Ethiopia, and Uganda, all low-GDP. Political stability matters here because it distinguishes Kenya and Ethiopia (relatively stable, large distance pipelines) from comparably-positioned but less stable peers.

Throws and jumps look weak. Modest population elasticity for throws (+0.18), nothing significant for jumps. Both fail to identify clean economic patterns at the 40-country panel size.

The country-level pooled result of Section ?? averages over these very different sport-specific models. A complete account of international NCAA recruitment would estimate event-specific gravity equations with sport-level demand-side covariates; we leave this to future work.

3.6 Region fixed effects horse-race

To quantify how much of the unexplained variation in country-panel supply is regional-pipeline-shaped, we add eight-region fixed effects to the count specification.

	Macro only	Macro + 8 region FE
$\log \text{pop}_{15-24}$	+0.35*** (0.07)	+0.40*** (0.07)
$\log \text{GDPpc}$	+0.37** (0.16)	+0.53*** (0.18)
Political stability	+0.38 (0.24)	+0.38 (0.24)
R^2	0.21	0.32

Adding region FE raises R^2 from 0.21 to 0.32, a meaningful but not dominant gain. A specification with region FE only (no macro covariates) achieves $R^2 = 0.13$. Read together: macro covariates

and regional pipelines each contribute, in roughly equal share, to a total explained variance of about a third. The other two-thirds remain unexplained even with both—consistent with the discussion in Section ?? that country-specific (not region-specific) pipeline structure dominates.

4 Discussion

Four substantive conclusions emerge.

First, **wealth and population scale govern entry into the international NCAA pipeline**. Both $\log \text{GDPpc}$ and $\log \text{pop}$ enter the extensive-margin probit at $p < 0.01$; political stability does not. The mechanism is the fixed-cost story familiar from migration economics: producing a recruitable athlete requires some minimum infrastructure and discovery-cost expenditure that scales with national resource flow.

Second, **conditional on entry, athletes scale sublinearly with cohort population**. The estimated elasticity is between 0.33 (OLS) and 0.61 (Heckman two-step), with the selection-corrected estimate being our preferred reading. Doubling a source-country’s 15–24 cohort raises NCAA supply by roughly $1.5\times$, not $2\times$. The most plausible mechanism is finite US-side recruitment capacity: the marginal NCAA program does not scale its international intake one-for-one with any single source country. Small wealthy countries are therefore over-represented per capita relative to populous ones. Quantile regression refines this: the elasticity is 0.24 at the low end of supply and 0.51 at the high end, so the diminishing-returns story is strongest for small-supply countries.

Third, **political stability has an intensive-margin effect of uncertain magnitude**. OLS gives a statistically zero estimate of $+0.37$ (SE 0.24), but with $n = 100$ the minimum detectable effect at 80% power is 0.68, so the data do not strongly reject moderately positive effects. The Negative Binomial estimate is $+0.79$ ($p < 0.01$); the Heckman estimate is $+0.60$ ($p < 0.10$). We cannot pin down whether stability matters at the country-pool level, because the country pool averages over very heterogeneous sport structures. *Within* distance running, however, political stability is large and significant ($+0.93$, $p < 0.05$): the East African pipeline depends on it. Within sprints, it does not. Within throws and jumps, sample sizes preclude a strong claim.

Fourth, **the relevant unit of analysis may not be the country**. Adding eight region fixed effects raises R^2 from 0.21 to 0.32 in the count specification; the macro covariates’ point estimates barely move. Region absorbs some but not most of the unexplained variance. The sport-event split is more informative than the regional split. Both point to the same conclusion: the country-panel pooled regression is averaging over qualitatively different supply curves (Caribbean sprint network, East African distance network, European throwing pipelines, etc.) that share little structure beyond GDP and population. A complete model would estimate gravity equations at the sport-event level with bilateral US-source-country variables (recruiting links, scholarship agreements, language commonality, alumni networks) that we do not have in our data.

5 Caveats and limitations

- **Cross-section, not panel.** Roster snapshots are 2024–25 / 2025–26; macro covariates are 2023 and 2010–23 aggregates. A descriptive pseudo-panel based on athletes’ class-year gives at most five entry cohorts with no clean pre-period for any shock event.

- **Coverage.** Hometown is observed for 87.6% of track athletes; the unobserved 12.4% are missing not at random (JS-rendered roster pages that systematically affect a few major D-I programs).
- **Track-and-field dominance.** 96% of the international sample competes in track. The paper’s findings should be read as describing the international track-and-field market specifically.
- **Cohort denominator approximation.** We use the average of the 15–24-year-old cohort 2018–2022 as the denominator. The ideal denominator would be 18–22 to match enrolled ages; we attempted this but the World Bank does not provide aggregate 18–22 totals. Rescaling 15–24 by a constant fraction is observationally equivalent in the count model (the constant is absorbed by the intercept), so this does not change our slope estimates.
- **Heckman identification by functional form.** The exclusion restriction in our Heckman two-step is the probit nonlinearity, not a covariate-level exclusion. The IMR coefficient (+3.27, SE 2.64) is large but imprecisely estimated, so selection-correction matters but its magnitude is uncertain. A proper exclusion (e.g. a US-side measure of recruitment intensity) would improve identification.
- **Modest R^2 .** Macro + region explain about a third of intensive-margin variance; the residual is most plausibly driven by country- and sport-specific recruitment pipelines that we cannot proxy.

6 Reproducibility

All results in this paper are reproducible from local Python scripts:

- Roster scraping: `discover_track_urls.py`, `run_track_scrape.py`, `run_track_selenium.py`, `rediscover_urls.py`.
- Country mapping: `run_country_mapping.py`.
- Macro merge: `run_merge_analysis.py`, `fetch_extra_shocks.py`, `run_join_shocks.py`.
- Main regressions: `run_main_results.py` (extensive probit + intensive OLS chain).
- `run_revisions_2.py`: Heckman two-step, MDE for political stability, quantile regression at $\tau \in \{0.25, 0.50, 0.75\}$, Negative Binomial with free log pop, within-event count regressions, and region FE horse-race — Tables 3, 4, 5, 6 of this paper.
- `run_stress_tests.py`: drop-top-5 disaster countries, events-vs-deaths comparison, refugees raw count.
- `run_bartik.py`, `run_event_study.py`, `run_paper_outputs.py`: auxiliary results referenced in Sections 3.6 and 5.

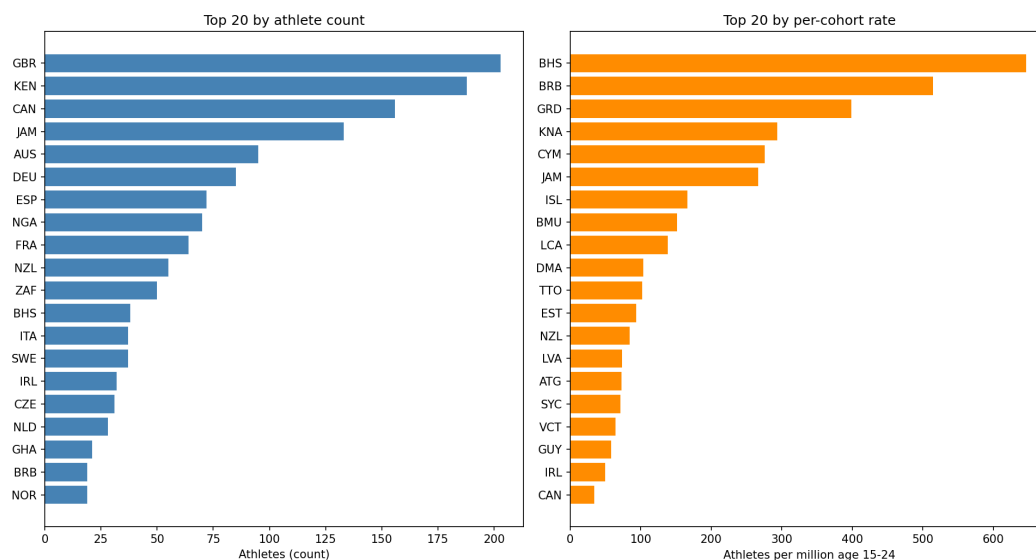


Figure 1: Top 20 source countries by raw athlete count (left) and by athletes per million 15–24-year-olds (right).

Figures

References

- Cohen, J. (1988). *Statistical Power Analysis for the Behavioral Sciences* (2nd ed.). Lawrence Erlbaum Associates.
- Lee, D. S., J. McCrary, M. Moreira, and J. Porter (2022). “Valid t -Ratio Inference for IV.” *American Economic Review* 112(10): 3260–3290.

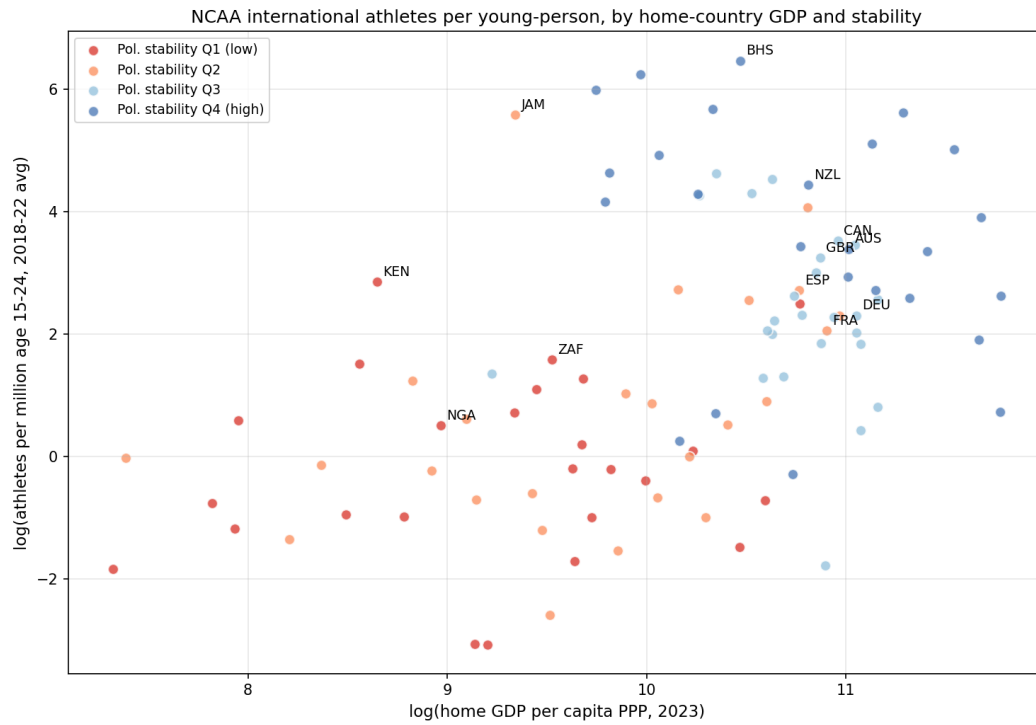


Figure 2: Log athletes per million age 15–24 vs. log home-country GDP per capita, colour-coded by political-stability quartile. The apparent stability gradient is largely a country-size artefact in this view; see Section ??.